

Residue Theorem and Applications

Complex Analysis

Residue Theorem

Let f be analytic inside and on a simple closed contour C except at isolated singularities $z_1, \dots, z_n$.

Then

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k)$$

Thus contour integrals reduce to computing residues.

Residue Shortcuts

If z_0 is a simple pole,

$$\operatorname{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0)f(z)$$

If

$$f(z) = \frac{p(z)}{q(z)}$$

and z_0 is a simple zero of $q(z)$ then

$$\operatorname{Res}(f, z_0) = \frac{p(z_0)}{q'(z_0)}$$

Pole of order m :

$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$$

Improper Integrals on $(-\infty, \infty)$

Suppose

$$I = \int_{-\infty}^{\infty} R(x) dx$$

with R rational.

Procedure:

1. Extend $R(x)$ to $R(z)$.
2. Integrate over a semicircle in the upper half-plane.
3. Apply the residue theorem.
4. Let $R \rightarrow \infty$.

Then

$$\int_{-\infty}^{\infty} R(x) dx = 2\pi i \sum \text{Res}(R, z_k)$$

for poles in the upper half-plane.

Example

$$\int_{-\infty}^{\infty} \frac{dx}{x^2 + 1}$$

Poles:

$$z = \pm i$$

Upper half-plane pole: $z = i$

Residue:

$$\text{Res} = \frac{1}{2i}$$

Result:

$$\int_{-\infty}^{\infty} \frac{dx}{x^2 + 1} = \pi$$

Section 8.4: Trigonometric Integrals

Many integrals have the form

$$\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta$$

Use

$$z = e^{i\theta}$$

Then

$$d\theta = \frac{dz}{iz}$$

and

$$\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$\sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right)$$

The integral becomes a contour integral around the unit circle.

Example

Evaluate

$$\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$$

Substitute

$$\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$d\theta = \frac{dz}{iz}$$

Then

$$I = \oint_{|z|=1} \frac{2}{z^2 + 4z + 1} dz$$

Poles

Solve

$$z^2 + 4z + 1 = 0$$

$$z = -2 \pm \sqrt{3}$$

Only

$$z = -2 + \sqrt{3}$$

lies inside $|z| < 1$.

Residue

For

$$f(z) = \frac{2}{z^2 + 4z + 1}$$

Residue formula:

$$\text{Res} = \frac{2}{2z + 4}$$

At $z = -2 + \sqrt{3}$:

$$\text{Res} = \frac{1}{\sqrt{3}}$$

Result

Residue theorem:

$$I = 2\pi i \left(\frac{1}{i\sqrt{3}} \right)$$

Thus

$$\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta} = \frac{2\pi}{\sqrt{3}}$$

General Example

Evaluate

$$I = \int_0^{2\pi} \frac{d\theta}{a + b \cos \theta}$$

with $a > |b|$.

Using

$$z = e^{i\theta}$$

the integral becomes a rational contour integral.

Poles occur where

$$z^2 + \frac{2a}{b}z + 1 = 0$$

One root lies inside the unit circle.

Result

Computing the residue gives

$$\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}$$

This classic result follows directly from residues.

Section 8.5: Indented Contour Integrals

Some real integrals contain poles on the real axis:

$$\int_{-\infty}^{\infty} \frac{f(x)}{x-a} dx$$

The integrand is singular at $x = a$.

Standard contour methods fail because the pole lies on the contour.

Indented Contours

We avoid the pole by indenting the contour around the singularity.
The contour consists of

- ▶ the real axis
- ▶ a small semicircle around the pole
- ▶ a large semicircle closing the contour

This allows the residue theorem to be applied.

Cauchy Principal Value

For integrals with singularities on the real axis we define

$$\text{PV} \int_{-\infty}^{\infty} f(x) dx = \lim_{\epsilon \rightarrow 0} \left(\int_{-\infty}^{a-\epsilon} f(x) dx + \int_{a+\epsilon}^{\infty} f(x) dx \right)$$

The indentation contributes half of the residue.

Key Result

If f has a simple pole on the real axis at a then

$$\text{PV} \int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \text{Res (upper half-plane)} + \pi i \text{Res}(f, a)$$

The pole on the contour contributes **half** a residue.

Summary

Residues allow us to evaluate

- ▶ rational integrals on $(-\infty, \infty)$
- ▶ trigonometric integrals
- ▶ integrals with real-axis singularities (principal values)

These methods form one of the most powerful applications of complex analysis.